MA.912.AR.3.6

Benchmark

MA.912.AR.3.6 **Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.**

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.1.3, MA.912.AR.1.7
- MA.912.F.1.6

Terms from the K-12 Glossary

- Quadratic Function
- x-intercept
- y-intercept

Vertical Alignment

Previous Benchmarks

- MA.8.AR.1.2

Next Benchmarks

- MA.912.AR.6.4, MA.912.AR.6.5

Purpose and Instructional Strategies

In grade 8, students multiplied two linear expressions to obtain a quadratic expression. In Algebra I, students transform a quadratic function to highlight and interpret its vertex or its zeroes. In later courses, students will determine key features of higher degree polynomials.

- Instruction includes making connections to various forms of quadratic equations to show their equivalency. Students should understand and interpret when one form might be more useful than another depending on the context.
    - Standard Form
      Can be described by the equation $y = ax^2 + bx + c$, where a, b and c are any rational number. This form can be useful when identifying the y-intercept.
    - Factored form
      Can be described by the equation $y = a(x - r_1)(x - r_2)$, where r_1 and r_2 are real numbers and the roots, or x-intercepts. This form can be useful when identifying the x-intercepts, or roots.
    - Vertex form
      Can be described by the equation $y = a(x - h)^2 + k$, where the point (h, k) is the vertex. This form can be useful when identifying the vertex, which is the turning point of the function where the axis of symmetry intersects.
- Instruction includes comparing and contrasting between a linear function of the form $y = a(x - h) + k$ and a quadratic function of the form $y = a(x - h)^2 + k$. This will also extend to an absolute value function of the form $y = a|x - h| + k$. *(MTR.5.1)*
- Instruction includes the use of x-y notation and function notation.

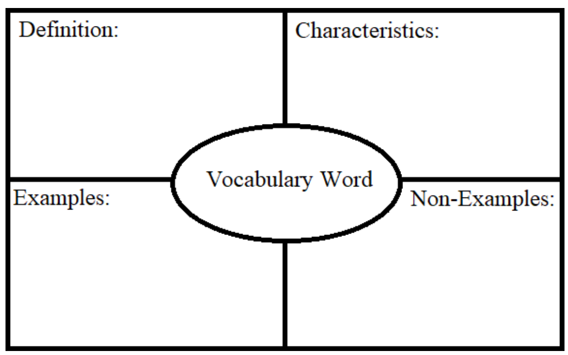
- Most contexts for this benchmark will present functions in standard form. Depending on their perspectives, students might take several approaches to determine vertices and zeros. Have students discuss strategies they might use. Let students know they should use the approach that is most efficient for them (*MTR.3.1*).
    - To determine zeros, students could use the quadratic formula or convert the function into factored form, or complete the square, or use Loh's method.
- To determine the vertex, students could convert the function into vertex form or determine the axis of symmetry ($x = \frac{-b}{2a}$). Calculate this value from one of the previous functions discussed and guide students to see that the vertex of each parabola falls on the line of symmetry. Considering this, they can substitute that x-value into the function to determine the corresponding y-value of the vertex.

Common Misconceptions or Errors

- Some students may have difficulty interpreting the meaning of the zeros and vertex.

Strategies to Support Tiered Instruction

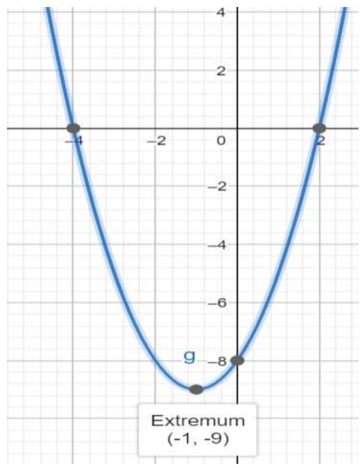
- Teacher provides a graphic organizer for key terms (zeros and vertex) that can be created using information provided in a given problem.



- Teacher co-creates a graphic organizer to compare real-world and mathematical contexts related to quadratics.
    - For example, the chart below can be used to compare mathematical terms with real-world context.

x		y	
Mathematical Context	**Real-world Context**	**Mathematical Context**	**Real-world Context**
Independent variable or input	Time	Dependent variable or output	Height
x-values, roots, solutions	Seconds for a ball to reach ground	y-value of vertex	Maximum height of a ball

- Teacher provides a visual aid, including using graphing software, to interpret the zeros and vertex of a quadratic function. Students can compare definitions and examples to determine the zeros and vertex of the parabola.
    - For example, the roots of the graph shown are (2, 0) and (−4, 0). The locations on the graph that intersect the x-axis are the roots. The roots are also the solutions of the quadratic. The vertex of the graph shown is $(-1, -9)$. The vertex is the minimum or maximum point (sometimes called the extrema) of the parabola. The vertex is also the point where the quadratic changes from decreasing to increasing or from increasing to decreasing, and is halfway between the roots.



Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.6.1, MTR.7.1)

A diver jumps off a cliff 5 meters high into a lake. The diver's position can be represented by the function $h(t) = -4.9t^2 + 1.5t + 5$, where h represents the diver's height relative to the lake's surface and t represents the time in seconds.

Part A. Determine the roots of the function described.
Part B. Interpret each with respect to the situation.

Instructional Task 2 (MTR.3.1, MTR.7.1)

A local campground charges \$23.50 per night per campsite. They average about 32 campsites rented each night. A recent survey indicated that for every \$0.50 decrease, the number of campsites rented increases by five.

Part A. Write a quadratic equation that describes this situation.
Part B. Determine the vertex and zeroes of the function described.
Part C. Interpret each with respect to the situation.
Part D. What price will maximize revenue?

Instructional Items

Instructional Item 1

Marcus just purchased a Super Bouncy Ball from a local toy store. Once outside, he throws it down to see how high the ball will bounce. The function $h(x) = -2x^2 + 24x - 31.5$ represents the height, h in inches, above the ground of the ball as it relates to its horizontal distance from Marcus x, in inches. Find the zeros and vertex of this function and interpret the meaning of each.

Instructional Item 2

A hallway has an arch that is in the shape of a parabola. The arch is modeled by the function $y = -5x^2 + 12x$, where x represents the horizontal measurement, in feet, and y represents the vertical measurement, in feet. What are the zeroes of this function, and what do they represent?

A. The zeroes are (2.4,0) and (1.2, 7.2), which means these are the two places the archway touches the ground.
B. The zeroes are (0,0) and (2.4, 0), which means these are the two places the archway touches the ground.
C. The zeroes are (0,0) and (1.2, 7.2), which means that the maximum height of the arch is 7.2 feet.
D. The zeroes are (2.4, 0) and (1.2, 7.2), which means that the arch is 7.2 feet wide at the base.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*